



Do workers maximize expected utility when choosing labour contracts under ambiguity?

Xueting Yang¹ · John Hey¹

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Abstract

Many labour contracts are (deliberately or necessarily) ambiguous in nature. This paper experimentally investigates the implications. We ran an experiment in which subjects, as workers, are given a choice between an ambiguous contract and a contract with a fixed and known income for carrying out a specific task. The task, which they are then asked to complete, is also part of the experiment, and we observe what they do. Ambiguity is a key part of the ambiguous contract – subjects know the possible payments but are not told the chances of getting them. To understand the results, we need to use one of the many theoretical models of decision-making under ambiguity. We chose a subset, namely: α MaxMin, MaxMin, Expected Utility (EU), and Rank Dependent Expected Utility (RDEU). The first two of these explicitly characterise ambiguity; the last two reduce ambiguity to risk. We fit each of these four models to our experimental data and rank the models according to the Akaike criterion. EU and RDEU emerges as the best. Whether this is because of the complexity of the problem, or something inherent remains to be discovered.

Keywords Ambiguity · Risk · Work · Experiments · Payment schemes · Allocation elicitation method

JEL codes D01 · C91 · D81 · J33

✉ John Hey
john.hey@york.ac.uk

Xueting Yang
xueting.yang@york.ac.uk

¹ Department of Economics and Related Studies, University of York,
Heslington, York YO105DD, UK

1 Introduction

We are interested in how workers react to the ambiguity built into some ambiguous labour contracts, this latter being defined as one in which the possible payments to the workers are known, but not their chances. For example, the worker is given a bonus if he/she completes 120% of some target. The worker does not know *ex ante* whether he/she will be able to meet this condition. To analyse the decisions that the subjects make during the experiment we need to choose a model that could explain their behaviour. As we are dealing with situations of ambiguity, it is natural to turn to models of behaviour under ambiguity. One obvious candidate that comes to mind is α Maxmin (Ghirardato et al., 2004), in which the decision-maker (DM) maximises a weighted average of the worst and best expected utility; another is *Maxmin* (Gilboa & Schmeidler, 1989), where the DM is extremely pessimistic and simply maximises the worst possible expected utility. We fit both these models to our data. We also fit Expected Utility (EU) theory, and Rank Dependent Expected Utility (RDEU). Our strategy is to see which of these four specifications (α Maxmin, Maxmin, EU and RDEU) best explains their behaviour. Based on the model fitting results, we can infer how a majority of DM processes ambiguity in the working context we built in the experiment.

After a brief introduction to related literature, we explain the structure of our experiment. It was in three separate parts: the Risk Part, the Ambiguity Part and the Work Part in that order. Before the subjects tackled the Work Part, they were asked to complete a Work Task, and they were given a chance to practice the Work Task. We explain later what the Work Task was. For each Part, we asked subjects to complete a number of decision tasks. These were all *allocation problems* - which we describe below. The data from these allow us to estimate key parameters which are: the risk-aversion parameter, the ambiguity-aversion parameter, and the number of tasks the subject thinks that he/she can solve.

2 Related literature

The related literature has two separate strands: (1) the labour decision over ambiguous contracts; (2) the experimental method we used.

- (1) The literature on labor economics is vast, so we focus on studies examining how ambiguity influences labor contract choices—specifically, whether workers prefer ambiguous or certain contracts. Three key papers in this domain are Hannan et al. (2005), Gonzales et al. (2020), and Burke et al. (2023), all of which investigate whether workers favor bonus contracts over penalty contracts.

Hannan et al. (2005) pioneered this line of inquiry by studying worker preferences between certain and risky contracts, as well as between bonus and penalty contracts. They found that employees generally prefer bonus contracts, though such contracts are less effective in motivating effort. Gonzales et al. (2020) extended this research by testing the phenomenon in a two-sided labor market where both employers and

workers make choices. In contrast, our study examines only one side of the market and does not incorporate the concept of a penalty contract.

Burke et al. (2023) further advanced this literature by comparing different *frames* of ambiguous contracts—penalty versus bonus. In their experiment, subjects in the *Unambiguous Payoffs* condition knew their exact incentive compensation, while those in the *Ambiguous Payoff* condition faced a range of possible outcomes without explicit probabilities. Our design differs in two key ways:

- a) Nature of Ambiguity: Burke et al. (2023) focus on ambiguity over outcomes, whereas we examine ambiguity over probabilities. Specifically, we specify two possible probability values rather than a range of outcomes.
- b) Decision Context: they study individual effort decisions under different contract frames, whereas we investigate allocation decisions between ambiguous and certain payment schemes, emphasizing how workers' ambiguity perceptions and attitudes (risk/ambiguity aversion) shape their choices.

In addition to points mentioned above, there are several key differences between our paper and theirs. First, we implement genuine ambiguity in our experiment. Second, we analyse our data subject by subject, rather than across subjects, thus getting an idea of the different types in our subject pool. Third, we provide stronger incentive compatibility to ensure truthful reporting. Fourth, we elicit individual parameters describing behaviour (using the Allocation method) rather than just asking (relatively uninformative) pairwise choice questions.

- (2) As to our choice of experimental method, we used the Allocation Method to elicit preferences (we describe this in detail below). This method is easier for subjects to understand compared with the Becker-DeGroot-Marschak (BDM) method (Becker et al., 1964), and avoids generating conflicted preferences which is what the multiple price list method (Holt & Laury, 2002) may cause if the lists are not designed carefully. Loomes (1991) was the pioneer of the Allocation method in which, given a certain endowment of tokens, subjects were asked to allocate it to different possibilities with corresponding chances. Carbone et al. (2017) applied this method to elicit ambiguity attitudes and investigated several theoretical modes to find which one best explains the experimental results. Zhou & Hey (2018) used various methods including the allocation method in risk preference elicitation, and their results showed that elicited individuals' risk attitudes may vary over different methods. The above literature can be classified as the stream of exploring the relationship between theories and experiments. Ahn et al. (2014) employed a financial market context and asked subjects to implement portfolio choice problems via allocation. Assets in portfolios were risky, but the number of assets and the probability of the portfolio happening were unknown. Apart from Ahn et al., few experimental researchers have investigated ambiguity preference in the labour market and used the allocation method. Our study fills this gap and focuses on the labour market, asking subjects to allocate payoffs over different payment schemes with real-effort work.

As for the work task – finding two elements in a matrix adding up to a targeted number – which we describe in detail below, is commonly used in studies to check work performance incorporating monetary incentives or reward structures, but fewer studies connected with ambiguity perception. Although this kind of task requires cognitive skills, it is familiar to subjects and easy to implement (Ariely et al., 2009). We note the research of Macera (2018) which set up two payment schemes in experiments with the work task: (a) current performance contributes to both present and future payments; (b) present effort supply only relates to future incentives; and found that deferring incentives would not negatively affect workers' effort provision. Macera explained this finding by highlighting the role of loss aversion based on the reference-dependent preferences theory proposed by Kőszegi & Rabin (2006), and stated that wage uncertainty boosts people to form a negative reference-dependent expected utility because of loss aversion. A fixed current payoff can pacify such pain brought by uncertainty. However, the role of uncertainty was not explored; it was only a possible reason he proposed.

We use the method applied by Macera to provide a work task in our experiment: in the work part, subjects were required to complete decision problems in which subjects were presented with one ambiguous lottery (the outcome of which depended on their performance in a work task to be completed later) and one certainty. The payoff they allocated to the ambiguous lottery is called the uncertain income. That allocated to the certainty is called the fixed income. Whether they can obtain the uncertain income depends on the number of puzzles they solve correctly in the work task.

3 The allocation method for eliciting parameters

The Allocation method is an accepted method for eliciting risk and ambiguity preferences. Let us describe it carefully since it is central to our experiment. We start by describing here the Allocation method as applied to eliciting risk preferences, with differences for Ambiguity and Work described later.

The method involves a series of decision problems, in each of which subjects are presented with two possibilities, called Red and Yellow, which have given stated chances of occurring, and are asked to allocate a given endowment of tokens E (always 100 in our experiment) between the two colours. For each colour, there is a given exchange rate between tokens and money; these are denoted by e_R and e_Y . A given allocation of tokens (AL to Red and $E-AL$ to Yellow) implies an allocation of money: $e_R AL$ to Red and $e_Y(E-AL)$ to Yellow. In the Risk part, both Red and Yellow are lotteries: Red is a lottery between the amount of money allocated to Red and £0, with stated chances; Yellow is a lottery between the amount of money allocated to Yellow and £0, with stated chances. To 'play out' a problem, the computer chooses Red or Yellow, with the stated chances. Then, if Red occurs, what happens is that the Red lottery is played out, with the payment being *either* $£e_R AL$ (the amount of money allocated to Red) *or* £0 (with the previously stated chances). If instead Yellow occurs, what happens is that the Yellow lottery is played out, with the payment being *either* $£e_Y(E-AL)$ (the amount of money allocated to Yellow) *or* £0 (with the previously stated chances).

A screen shot of the experimental interface is shown in Fig. 1 below:

This figure is an example of a Risk case and shows the implications of an equal (50:50) allocation of tokens. *With such an equal allocation*, given the exchange rates between tokens and money, the allocation implies £8.25 to Red and £7.50 to Yellow. Here, Red and Yellow have chances 70% and 30% of happening. If Red is chosen by the computer, a risky lottery between £8.25 with chance 70% and £0 with the residual chance is played out; If Yellow is chosen by the computer, a risky lottery between £7.50 with chance 30% and £0 with the residual chance is played out. Overall, then, reducing the two stages to one, the problem gives £8.25 with chance 49%, £7.50 with chance 9% and £0 with the residual chance. *It is crucial to note, as was made clear to our subjects, that changing the allocation does not change the chances*, but only the amounts paid out: for example, an allocation of all 100 tokens to Red, the problem gives £16.50 with chance 49%, £0 with chance 9% and £0 with the residual chance; an allocation of all 100 tokens to Yellow, the problem gives £0.00 with chance 49%, £15.00 with chance 9% and £0 with the residual chance.

Subjects could explore the implications of different allocations by moving the slider at the bottom left. The graph on the right shows the implications: here, with 50:50 allocations, the Red rectangle is of height £8.25 and width 49%; the Yellow rectangle is of height £7.50 and width 9%. As they moved the slider, they could see that the *heights* but not the *widths* changed. It should be noted that the choice of an allocation indicates the *attitude to risk* of the subject, though this depends upon the two lotteries.

In the Ambiguity part, we again used the Allocation method, but here Red implied an ambiguous (rather than a risky) lottery – in which the chance of them being paid the amount of money allocated to Red was not told to them precisely; instead, *two possible values for the chance were given* – one the lowest chance, the other the greatest chance. Here, the choice of an allocation indicates the *attitude to ambiguity*

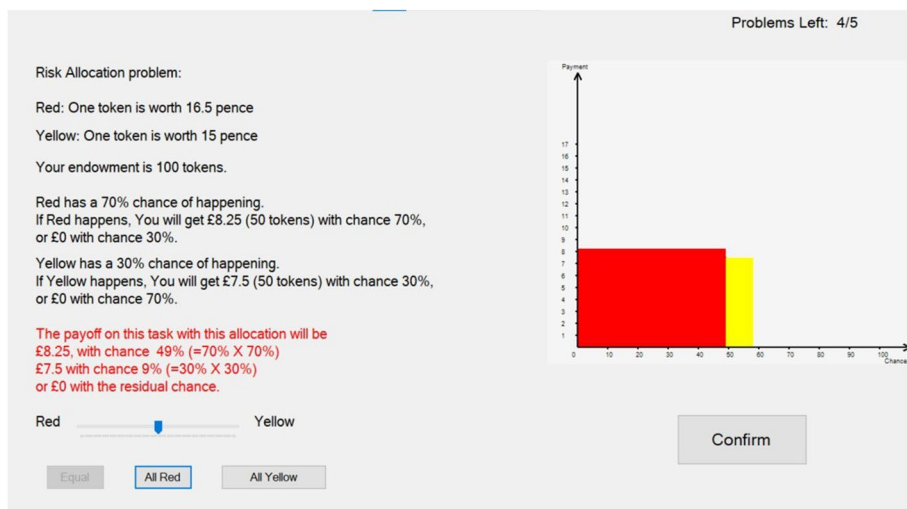


Fig. 1 Experimental interface for Allocation problems in the risk part

of the subject. If subjects put all their endowment on Yellow, that reveals that they are averse to ambiguity. If they put all their endowment on Red, that reveals that they prefer ambiguity.

In the Work part, we again used the Allocation method, but here Red implied an ambiguous (rather than a risky) lottery – in which the possibility of being paid the amount of money allocated to Red depended upon their future performance in the Work Task; this latter is clearly in general ambiguous. In contrast, Yellow implied a certain income. Here the choice of an allocation indicates the *perception of ambiguity* of the subject. We discuss this point further in the Experimental Design section.

4 Experimental design

The experiment was conducted at the University of York with 5 sessions and 94 participants in total, including undergraduate and postgraduate students and staffs¹ from different departments. Each session lasted around 1.50 h. Subjects' payments depended on the playing out of a randomly chosen problem in each part and their performance in the Work Task. They were paid the sum of these payments plus a £2.50 show-up fee after finishing the experiment. The average payment was £17.10 per subject.

In summary, we implemented a real-effort laboratory experiment consisting of three parts: a Risk part, an Ambiguity part, and a Work part in that order. In each part, subjects were asked to make decisions on allocation problems. After each part, one of their decisions would be played out by the computer randomly, and the corresponding payoff would be their payment for that part.

Before the experiment, we calibrated the problems: we carried out a *simulation* via MATLAB to determine the parameter sets and number of problems for each decision part. Given specified values of the parameters of the model, we applied the Maximum Likelihood method based on the three models to check whether the means of estimated results were close to these assumed values with a small variance over different sets of parameters.

In the Risk part, subjects were asked to allocate the endowment E , 100 tokens, between two risky lotteries: The Red Lottery (the money allocated to Red and £0 with stated chances) and the Yellow Lottery (the money allocated to Yellow and £0 with stated chances). There were 5 problems. The chance of Red happening and that of Yellow happening changed over problems, but not within a problem, and the exchange rate for tokens allocated to the Red lottery also changed over problems, but not within a problem. The software interface of this part is displayed in Fig. 1. The setting of the ambiguity part is similar to that of the risk part, but the Red Lottery was ambiguous. Subjects *were only informed of the lowest and highest chance* of getting the money allocated to Red if Red is chosen by the computer, instead of a specific

¹ As it is a real-effort experiment, both students and staff were treated as workers who were requested to do the work task, which may only involve basic mathematical calculation without other special techniques, so we think the identification difference between students and staff would not influence the results.

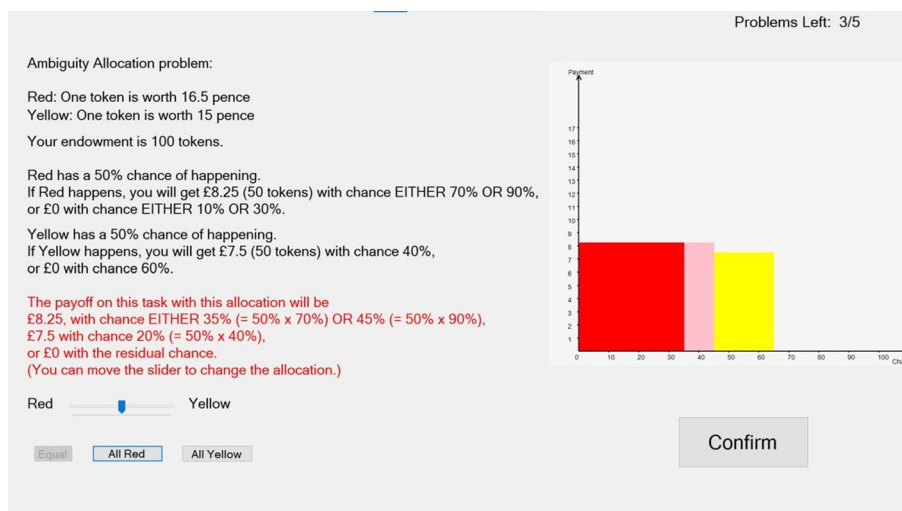


Fig. 2 Experimental interface for Allocation problem in the ambiguity part (Pink represents the ambiguous payoff)

value. This part also included 5 problems. Figure 2 shows the interface for ambiguous decision problems. Again, the problems were chosen after extensive simulation.

As for the Work part, it not only included 30 allocation problems (again chosen after extensive simulation), and also introduced a real-effort Work Task that consisted of 30 puzzles. Each puzzle consists of a 3 by 4 matrix of numbers which could be one-decimal, two-decimal, or three-decimal. Subjects were asked to find the unique pair of numbers adding up to 100, with a decreasing time constraint on each puzzle (example see Fig. 3). If subjects did not solve a puzzle within the time constraint, it would be counted as incorrect. The difficulty of a matrix might be influenced by the position of two correct numbers and decimal places of elements: if the correct numbers are closer, it might be easier for subjects to solve; It might take longer time to solve puzzles with three-decimal number, and although subjects might have the abil-



Fig. 3 Experimental Interface for Work Task

ity to find the correct answer they have already run out of time. The numbers in each matrix were assigned randomly by the computer, and thus the difficulty of the given matrix was also changed randomly.

In the Work Part, subjects were offered a practice round for the Work Task that was exactly the same as in the real Work task, except for the numbers in the matrices. This setting helps subjects build a rough idea about how many puzzles they can solve correctly. Then, they were asked to answer allocation decision problems that employed one ambiguous lottery and one certain outcome. The payoff they allocated to the ambiguous lottery we called the *uncertain income*, and that they allocated to the certain lottery we called the *fixed income*. In addition to the exchange rates of tokens and the chance of the ambiguous lottery happening, the required number of puzzles to be solved correctly also changed over problems. Whether they can get the uncertain income depends on the number of puzzles they solve correctly in the Work Task and whether Red is chosen by the computer. According to Fig. 4, for example, if a subject allocates 50 tokens with the exchange rate: 1 token = 10.5 pence to (Red (the ambiguous lottery in the experiment), pays £5.25 ($= 50 \times 10.5$ pence) if Red is chosen by the computer and the subject solves at least 18 puzzles correctly. Otherwise, he/she would get nothing. If Yellow is chosen by the computer, the subject gets paid £2.50 ($= (50 \times 10.5/100)$ pence) for sure. After completing the decision allocation problems, subjects tackled the real Work Task. Finally, the computer played out a randomly chosen one of the work allocation problems, with the payment to the subject depending on the number of puzzles that that subject had solved correctly.

We should discuss why we had three allocation parts. The reason is that we wanted to estimate key parameters: namely each subject's risk aversion for EU, r , and their parameter, g , of the weighting function for RDEU, their ambiguity aversion, α , for α MaxMin, and the number of problems that the subjects thought that they could solve in the Work Task. In order to estimate the latter, we need to know their ambiguity aver-

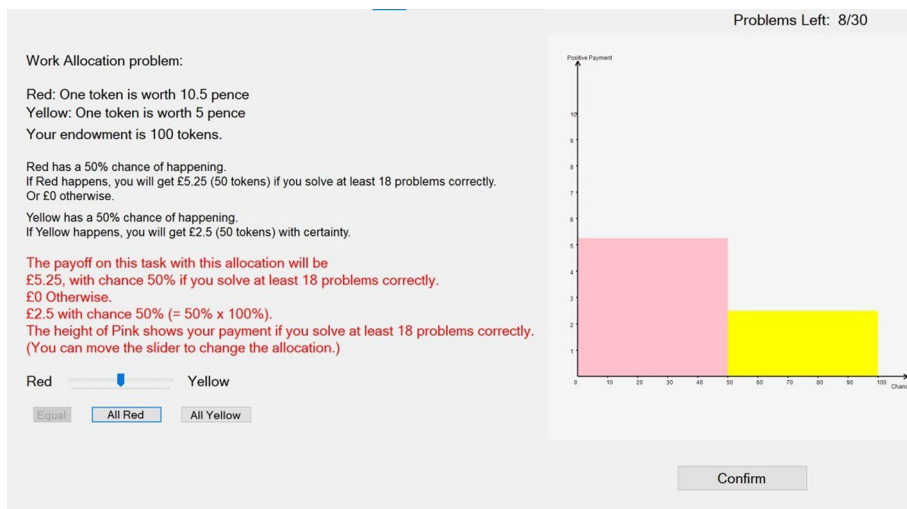


Fig. 4 Experimental interface for Allocation problem in the work case (Pink represents the ambiguous payoff)

sion, and in order to estimate their ambiguity aversion we needed to know their risk aversion. Identifiability issues stopped us from estimating all three simultaneously.

To summarise, we estimated the parameters r and np for EU; r , α , np and NP for α MaxMin; r and np for MaxMin; and r , g and np for RDEU. In addition, we have to specify the stochastics. Here we adopted the common Stochastic Decision Rule which says that, in each problem, the actual allocation is equal to the optimal allocation plus normally distributed noise with mean 0 and standard deviation σ . We assumed and estimated different σ 's for different sets of decision problems. In sum, therefore we estimated 4 parameters for EU: r , σr , σw and np ; 7 parameters for α MaxMin: r , α , σr , $\sigma \alpha$, σw , np and NP ; 4 parameters for MaxMin: r , σr , σw and np ; 5 parameters for RDEU: r , g , σr , σw and np .

5 Results

The detailed results are in a Data Appendix available on the EXEC website. These contain estimates of the parameters for each model and the corresponding Log Likelihoods. We start with the bottom line. For each model, we computed the Akaike Criterion (AIC). This is given by $AIC = 2k - 2\text{Loglikelihood}$, where k is the number of parameters of the model and Loglikelihood is the maximised log-likelihood². We then used the AIC to rank the four models. We got the following results (we had a total of 94 subjects): (Table 1).

Hence, EU emerges as the best model in explaining the data, with MaxMin and RDEU as close seconds, and α MaxMin as being an outlier (possibly because of its large number of parameters). It seems that the subjects are treating the ambiguity as being risk (just treating the minimum number of puzzles solved as being the number solved). This conclusion is reinforced by the histogram below of the estimated ambiguity aversion in the α MaxMin estimation, where it will be seen that many of the subjects are either completely pessimistic ($\alpha = 1$) or completely optimistic ($\alpha = 0$). (Fig. 5 and 6).

Such extremes do not appear in the estimates of risk aversion obtained from the risk part of the experiment:

As noted above, RDEU comes second in the rankings and the estimates of the parameter of the weighting function are reasonable:

Table 1 AIC ranking result of models

	best	second best	third best	worst
α MaxMin	5	3	9	77
MaxMin	21	30	36	7
EU	46	23	21	4
RDEU	22	38	28	6

² We took the maximised Log-Likelihoods for the combined risk and ambiguity parts (for EU, MaxMin and RDEU) or the separate risk and ambiguity parts (for α MaxMin) plus the maximised Log-Likelihood for the work part (for all 4 models), thus getting a single Log-Likelihood for each model.

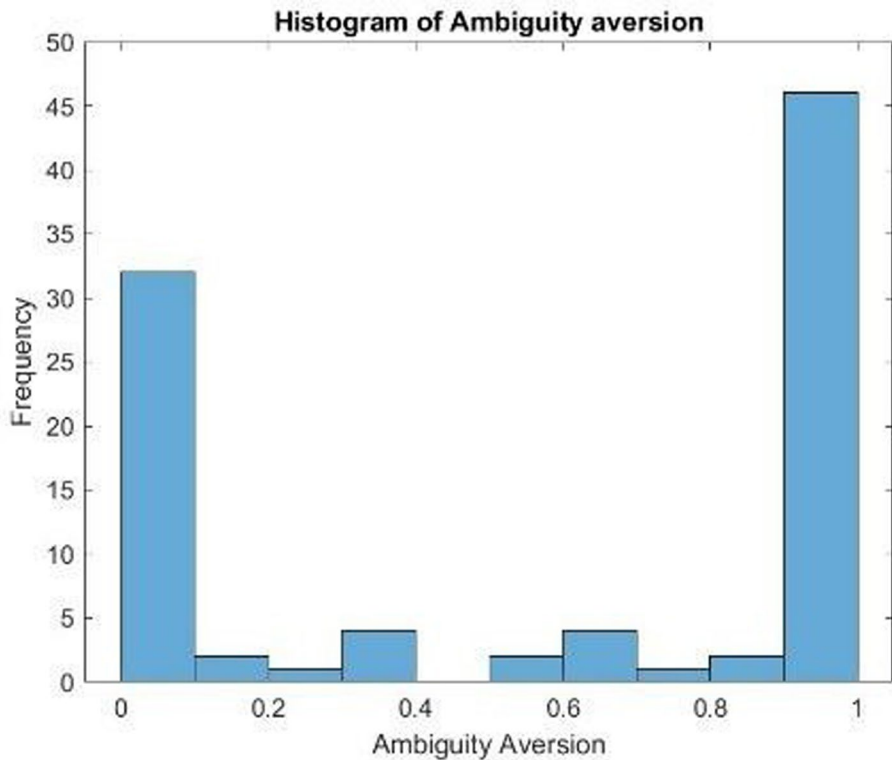


Fig. 5 estimated α under the α MaxMin model

To give some feel for what these values imply, plotted below are the implied weighting functions for the smallest and largest estimated values of the parameter. For the smallest value, the function has the typical shape: distorting low probabilities upwards and high probabilities downwards. For the largest value, the function distorts all probabilities downwards. When $g=1$, there is no probability distortion. As can be seen from the histogram above, most subjects distorted all probabilities downwards. (Fig. 7 and 8).

As EU seems to be the best model, it is interesting to explore it further. To check the reliability of the estimates, we look at the relationship between the estimated np (the subjects' perceptions of how many puzzles they think they can solve correctly), and the number they actually later solve correctly (*creal*). Regressing the latter on the former (using the estimated np from the EU model), we obtain (t-stat in parentheses).

$$creal = 0.79249 \, np(30.883) \quad (1)$$

94 observations, 93 error degrees of freedom, root mean squared error=6.68

It appears that subjects were rather good at predicting how many puzzles they would solve later.

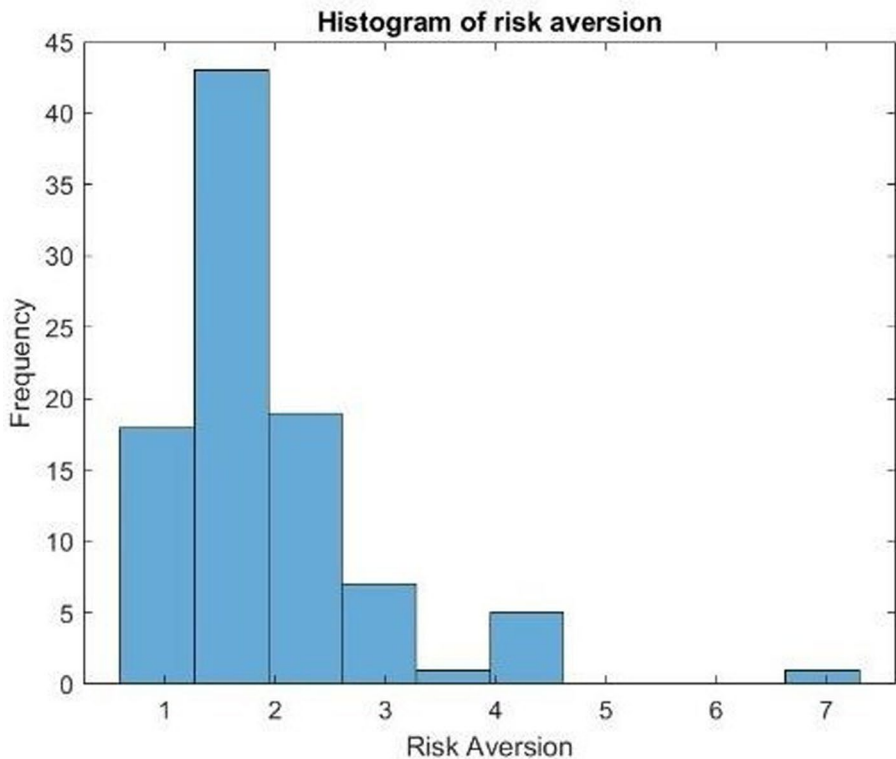


Fig. 6 estimated r under the α MaxMin model

Moreover, it is interesting to check whether they take the practice puzzles seriously. Regressing the number of correctly solved real puzzles (*creal*) against the number of correctly solved practice puzzles (*cprac*) we obtain (t-stat in parentheses):

$$creal = 1.1048 \ cprac(42.432) \quad (2)$$

94 observations, 93 error degrees of freedom, root mean squared error=4.96
creal and *cprac* are very close.

It seems to be the case that most subjects treated the practice puzzles seriously and they tried to build a sense or even use the practice result as a benchmark (like *np*) when they made decisions in work allocation problems. This result shows that individuals rely on their experience with the practice tasks to predict their future success with the real puzzles.

We should explain in more detail. Let us consider a subject who uses EU to solve the allocation work problems. Such a subject effectively reduces a subject's perception of ambiguity to knowledge of the number of puzzles that he/she thinks that he/she will be able to solve; this has been denoted by *np*. This clearly affects the allocation decision: *ceteris paribus*, the greater is *np*, the larger will be the allocation to Red (as the Technical Appendix makes clear). This is the reason that the maximum

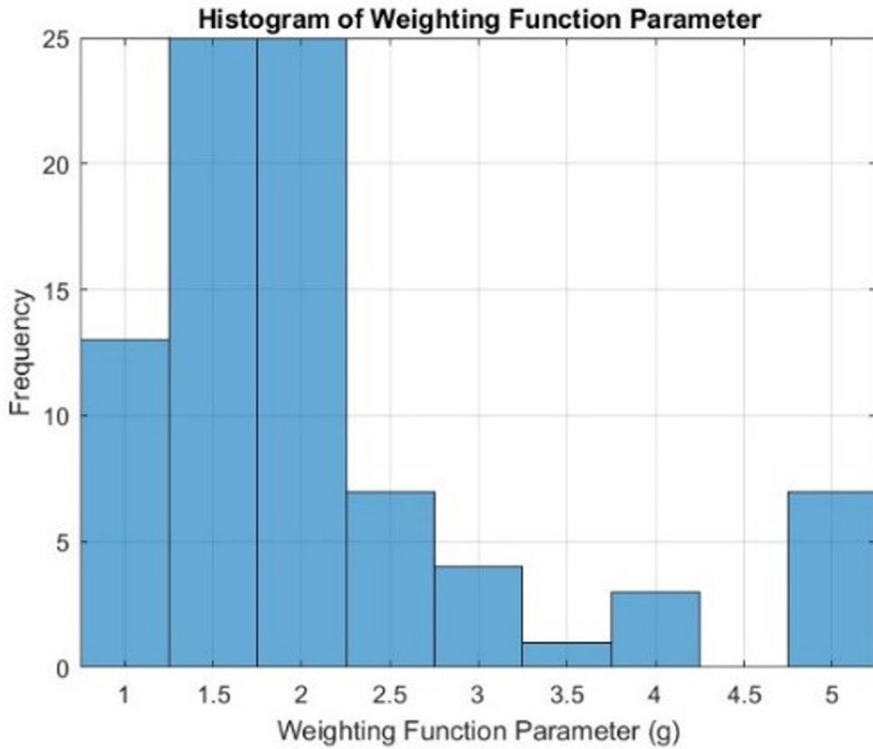


Fig. 7 estimated g under the RDEU model

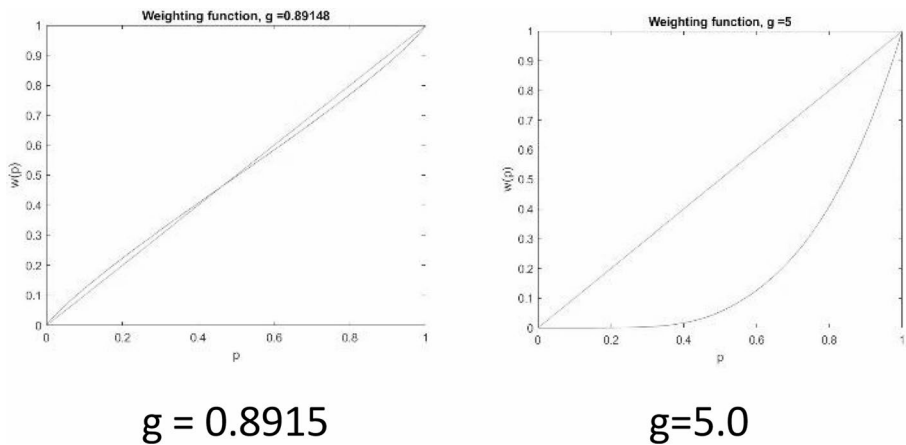


Fig. 8 Examples of how the value of g affects probability distortion

likelihood estimation enables us to estimate for each subject their value of np . We have shown that the estimated np (assuming EU) and the number of real puzzles solved correctly are closely related. Hence, we have shown that (assuming EU) a subject's perception of the ambiguity determines their work effort. If a subject uses one of the other models, the estimated value of np is less closely related to their work performance. This is shown by regressions similar to Eq. (1) above.

For RDEU we have a result quite close to that for EU:

$$creal = 0.82411 \ np(23.367)$$

94 observations, 93 error degrees of freedom, root mean squared error=8.54

For MaxMin we have a result further away from that for EU:

$$creal = 0.81499 \ np(30.389)$$

94 observations, 93 error degrees of freedom, root mean squared error=6.77

For α MaxMin we have a result quite a distance away from that for EU:

$$creal = 1.1467 \ np(19.582)$$

94 observations, 93 error degrees of freedom, root mean squared error=9.89

These regression results indicate (in addition to the ranking analysis) that these other models do not describe behaviour as well as EU, and confirms the ranking: EU is best, followed by RDEU, followed by Maxmin and a rather long way behind is α MaxMin. It could be argued that α MaxMin is the most 'ambiguity-oriented' model that we have considered, and hence that these results reinforce our conclusion that subjects were interpreting the problem as one of risk rather than ambiguity.

6 Conclusions

Our ambition with this experiment was to understand workers' perception of, and preference for, payment schemes that are ambiguous, and to determine whether their perception affects the work that workers do. Our results show that the majority of our subjects reduce the ambiguity to risk and then use EU to make decisions. Moreover, when we look closely at EU, we see that the subjects' estimates of how many puzzles they can solve (the amount of work that they think they will be able to do) is closely aligned to the actual number that they solve: EU seems internally consistent. Although subjects knew that the difficulty of each puzzle was random, their experience with the practice puzzles enabled them to perceive the ambiguity as risk.

What do these results imply about the way ahead - a future research agenda? One possibility is that the work context could be broadened, and the ambiguity made more obvious. Perhaps the 'playing out' of ambiguity could be done using the Bingo Blower or the Ambiguity Box. But, in the meantime we have answered the question in the title: the answer is yes.

Technical Appendix

This Appendix derives the optimal allocations for all cases and all models. First, a note on our methodology.

Methodology

We first used the Risk Data to estimate the CRRA risk aversion index r for each subject; then we combined this using the Ambiguity Data to estimate the ambiguity aversion α of the α MMD model for each subject. Then we combined this using the Work Data to estimate a range of how many puzzles each subject can solve correctly $[np, NP]$. We note that we have to proceed sequentially because of identifiability considerations.

Our estimation methodology in all three parts was the same. As the material below makes clear, the optimal allocation in each part depends upon the parameters we wish to estimate (as well as the parameters of the various problems). We assumed that the actual allocation in each case was equal to the optimal allocation plus normally distributed noise (with mean 0 and standard deviation σ). We then used maximum likelihood estimation to estimate the parameters we wish to estimate and σ . In addition to the estimated parameters, we obtained the maximised log-likelihood – a measure of the goodness-of-fit of that specification.

Optimal allocations

We start with a general result that we can apply to all the cases.

Let me start with the notation. I am going not to put an asterisk on the optimal allocation.

The Endowment: E ; Allocation to Red: AL ; Exchange Rate of Red: eR ; Exchange Rate of Yellow: eY ;

There are two outcomes for the Red Lottery: XR (the amount of money allocated to Red) with Probability p_{XR} and YR (always 0) with Probability $1 - p_{XR}$;

There are two outcomes for the Yellow Lottery: XY (the amount of money allocated to Yellow) with Probability p_{XY} and YR (always 0) with Probability $1 - p_{XY}$;

The probability of Red Lottery happening is: P_R ;

The probability of Yellow Lottery happening is: P_Y ;

So, the probability of getting the payoff XR is: $P_R p_{XR}$;

The probability of getting the payoff XY is: $P_Y p_{XY}$;

The general objective function takes one of the two forms below:

Case 1:

$$Au(eR \cdot AL, R) + Bu(eY \cdot (E - AL), R)$$

Or

Case 2:

$$Au(eY \cdot (E - AL), R) + Bu(eR \cdot AL, R)$$

Where A and B take specific forms which depend upon the probabilities and the case R is the CRRA index of risk aversion and u takes the form

$$u(x, R) = \frac{x^{1-R}}{1-R}$$

It follows that:

$$u'(x, R) = x^{-R}$$

Let us first do Case 1:

Differentiating the objective function w.r.t AL and putting the derivative equal to zero gives us

$$A \cdot eR \cdot u'(eR \cdot AL, R) = B \cdot eY \cdot u'(eY(E - AL), R)$$

Using $u'(x, R) = x^{-R}$ we get

$$A \cdot eR \cdot (eR \cdot AL)^{-R} = B \cdot eY \cdot (eY(E - AL))^{-R}$$

Now, taking the $-1/R^{th}$ root of each side, we get

$$(A \cdot eR)^{-1/R} \cdot eR \cdot AL = (B \cdot eY)^{-1/R} \cdot eY(E - AL)$$

Solving this for AL gives us

$$AL = \frac{EB^{-1/R}eY^{1-1/R}}{A^{-1/R}eR^{1-1/R} + B^{-1/R}eY^{1-1/R}} \quad (A1)$$

Now consider Case 2:

Now A and B take specific values depending upon the type

Differentiating the objective function w.r.t AL and putting the derivative equal to zero gives us

$$A \cdot eY \cdot u'(eY(E - AL), R) = B \cdot eR \cdot u'(eR \cdot AL, R)$$

Using $u'(x, R) = x^{-R}$ we get

$$A \cdot eY \cdot (eY(E - AL))^{-R} = B \cdot eR \cdot (eR \cdot AL)^{-R}$$

Now, taking the $-1/R^{th}$ root of each side, we get

$$(A \cdot eY)^{-1/R} \cdot eY(E - AL) = (B \cdot eR)^{-1/R} \cdot eR \cdot AL$$

Solving this for AL gives us

$$AL = \frac{EA^{-1/R}eY^{1-1/R}}{A^{-1/R}eY^{1-1/R} + B^{-1/R}eR^{1-1/R}} \quad (A2)$$

Comparing equations (1) and (2) we see that eR and eY are interchanged but so are A and B . Whether cases 1 and 2 are different depends on A and B and the various other items.

Now, let us apply this to the various cases. These are listed below.

Now, let us apply this to the various cases. These are listed below. Note the following special case: if $A=0$ in case 1 and if $B=0$ in case 2, the solutions give us $AL=E$, which is surely wrong. Just looking at the objective function tells us that AL must equal 0. So why does the solution go wrong? We think because 0 to the power of any negative number is infinite. So, we will take, when $A=0$ in case 1 and $B=0$ in case 2 that the optimal is $AL=0$.

In the following order, we consider the following models and cases:

EU Risk; EU Ambiguity and EU work

AMM Risk; AMM Ambiguity and AMM work

MM Risk; MM Ambiguity and MM work

RDEU Risk; RDEU Ambiguity and RDEU work

Let us work through these one by one:

EU risk

Here, for Case 1, $A=P_R p_{XR}$ and $B=P_Y p_{XY}$.

While for Case 2, $A=P_Y p_{XY}$ and $B=P_R p_{XR}$.

So, the case 1 optimum is $AL = \frac{E(P_Y p_{XY})^{-1/R} eY^{1-1/R}}{(P_R p_{XR})^{-1/R} eR^{1-1/R} + (P_Y p_{XY})^{-1/R} eY^{1-1/R}}$ (Opt EU Risk).

While the case 2 optimum is $AL = \frac{E(P_Y p_{XY})^{-1/R} eY^{1-1/R}}{(P_Y p_{XY})^{-1/R} eY^{1-1/R} + (P_R p_{XR})^{-1/R} eR^{1-1/R}}$ (Opt EU Risk).

I note that these are the same, so with EU **we do not need to consider cases 1 and 2 separately.**

EU ambiguity

An EU person just works with the simple average of the two Red positive payoff probabilities.

So $p_{XR} = \frac{p_{lXR} + p_{hXR}}{2}$ and all the rest is the same. So

$AL = \frac{E(P_Y p_{XY})^{-1/R} e^{Y^{1-1/R}}}{(P_R p_{XR})^{-1/R} e^{R^{1-1/R}} + (P_Y p_{XY})^{-1/R} e^{Y^{1-1/R}}}$ with $p_{XR} = \frac{pl_{XR} + ph_{XR}}{2}$ (Opt EU Ambiguity).

EU work Let me denote the number of puzzles needed by pn . Let me suppose that the subject has an upper bound on the number of puzzles that he/she can solve.

Let me denote the highest number by np . Then we have the following cases: 1.

If $pn \leq np$: So the probability, given R , of getting the large payment is p_{XR} .

And so, $A = P_R p_{XR}$ and $B = P_Y p_{XY}$ and we can use either Case 1 or Case 2 (proof above).

2. If $pn > np$, the optimal option is always allocating all tokens to Yellow, so: $AL^* = 0$. So, the optimal allocation is

$$AL = \frac{E(P_Y p_{XY})^{-1/R} e^{Y^{1-1/R}}}{(P_R p_{XR})^{-1/R} e^{R^{1-1/R}} + (P_Y p_{XY})^{-1/R} e^{Y^{1-1/R}}} \text{ if } pn \leq np \text{ Opt EU work}$$

0 if $pn > np$

AMM risk This is the same as EU risk:

$$AL = \frac{E(P_Y p_{XY})^{-1/R} e^{Y^{1-1/R}}}{(P_R p_{XR})^{-1/R} e^{R^{1-1/R}} + (P_Y p_{XY})^{-1/R} e^{Y^{1-1/R}}} \text{ (Opt AMM Risk)}$$

AMM ambiguity Here $A = \alpha P_R pl_{XR} + (1 - \alpha) P_R ph_{XR}$ and $B = P_Y p_{XY}$.

So, assuming Cases 1 and 2 give the same result we have.

$$AL = \frac{EB^{-1/R} e^{Y^{1-1/R}}}{A^{-1/R} e^{R^{1-1/R}} + B^{-1/R} e^{Y^{1-1/R}}} \text{ with } A = \alpha P_R pl_{XR} + (1 - \alpha) P_R ph_{XR} \text{ and } B = P_Y p_{XY} \text{ (Opt AMM Ambiguity)}$$

AMM work For the work case, $p_{XY} = 1$, and I need to specify the number of puzzles needed. I do not need to have pl_{XR} and ph_{XR} - just p_{XR} . They are replaced by the material below. Let me denote the number of puzzles needed by pn . Let me suppose that the subject has a lower and upper bound on the number of puzzles that he/she can solve. Let me denote the lowest number by np_1 and the highest number by np_2 , where $np_1 < np_2$. Then we have the following cases: 1. **I f**

$pn \leq np_1 < np_2$. The subject is in no doubt. Then the optimal allocation is given as in the EU risk case.

2. **If** $np_1 < pn \leq np_2$. Then the worst that can happen is that he/she solves none, $p_{XR} = 0$, and the best is that he/she solves pn , $p_{XR} = 1$. So, he/she gets $u(eR * AL)$ with probability (worst) 0 or P_R (best), and gets $u(eY * (E - AL))$ with probability P_Y

So, he/she as the same objective function as in Case 1, with $A = (1 - \alpha) P_R ph_{XR}$ and $B = P_Y$, and thus the solution is

$$AL = \frac{EB^{-1/R} e^{Y^{1-1/R}}}{A^{-1/R} e^{R^{1-1/R}} + B^{-1/R} e^{Y^{1-1/R}}} \text{ with } A = (1 - \alpha) P_R ph_{XR} \text{ and } B = P_Y \text{ (Opt AMM Work).3.}$$

If $np_1 < np_2 < pn$ Then the worst that can happen is that he/she solves none, and the best is that he/she solves *none*. So surely it is the best to put all on Yellow? Hence: $AL = 0$;

MM risk This surely must be the same as EU.

$$AL = \frac{E(P_Y p_{XY})^{-1/R} e^{Y^{1-1/R}}}{(P_R p_{XR})^{-1/R} e^{R^{1-1/R}} + (P_Y p_{XY})^{-1/R} e^{Y^{1-1/R}}} \text{ (Opt MM risk)}$$

MM ambiguity (we just put $\alpha=1$) Here we have $A = P_R p_{XR}$ and $B = P_Y p_{XY}$

So, the optimal allocation is

$$AL = \frac{EB^{-1/R} e^{Y^{1-1/R}}}{A^{-1/R} e^{R^{1-1/R}} + B^{-1/R} e^{Y^{1-1/R}}} \text{ with } A = P_R p_{XR} \text{ and } B = P_Y p_{XY} \text{ (Opt MM Ambiguity)}$$

MM work For the work case, $p_{XY}=1$, and I need to specify the number of puzzles needed. Let me denote the number of puzzles needed by pn . Let me suppose that the subject has a lower and upper bound on the number of puzzles that he/she can solve. Let me denote the lowest number by np_1 and the highest number by np_2 , where $np_1 < np_2$. Then we have the following cases: 1. **If $pn \leq np_1 < np_2$.** The subject is in no doubt. Then the optimal allocation is given as in the EU risk case.

2. **If $np_1 < pn \leq np_2$.** Then the worst that can happen is that he/she solves none, $p_{XR}=0$, and the best is that he/she solves pn , $p_{XR}=1$. So, he/she gets $u(eR*AL)$ with probability (worst) 0, and gets $u(eY*(E-AL))$ with probability P_Y . Then, he/she has the same objective function as in Case 1, with $A=0$ and $B=P_Y$, so the solution is

$$AL = \frac{EB^{-1/R} e^{Y^{1-1/R}}}{A^{-1/R} e^{R^{1-1/R}} + B^{-1/R} e^{Y^{1-1/R}}} \text{ with } A=0 \text{ and } B=P_Y \text{ (Opt MM Work)}$$

That is $AL=E$, but that cannot be true, for reasons given before. Surely $AL=0$. Yes, but this is a special case.

3. **If $np_1 < np_2 < pn$** Then the worst that can happen is that he/she solves none, and the best is that he/she solves *none*. So surely it is the best to put all on Yellow? Hence: $AL=0$;

RDEU risk In the case of **RDEU**, we assume for exposition that there are 3 outcomes O_1 , O_2 and O_3 with probabilities p_1 , p_2 and p_3 where the worst, O_1 , is ≤ 0 , where $O_1 < O_2 < O_3$ and $u(O_1)=0$ (Ranking from the worst to best):

$$\begin{aligned} REDU &= U(O_1)w(p_1) + U(O_2)(w(p_1 + p_2) - w(p_1)) + U(O_3)(1 - w(p_1 + p_2)) \\ &= U(O_2)(w(p_1 + p_2) - w(p_1)) + U(O_3)(1 - w(p_1 + p_2)) \\ w(p) &= \frac{p^g}{[p^g + (1 - p)^g]^{\frac{1}{g}}} \end{aligned}$$

The typical weighting function we used here is Tversky & Kahneman's inverse S-shaped function (Tversky & Kahneman, 1992).

There are two cases that we need to consider here because the payoff of Red is not always higher than that of Yellow.

Case 1:

$$(E - AL)eY \leq AL \cdot eR$$

(the non-negative payoff of Red is greater or equal to that of Yellow)

The objective function is of the form of equation 1 with

$$A = (1 - w(1 - P_R p_{XR})) \text{ and } B = (w(1 - P_R p_{XR}) - w(1 - P_R p_{XR} - P_Y p_{XY}))$$

So, the solution is $AL = \frac{EB^{-1/R}eY^{1-1/R}}{A^{-1/R}eR^{1-1/R} + B^{-1/R}eY^{1-1/R}}$ with A and B as above.

Case 2:

$$(E - AL)eY > AL \cdot eR$$

(the non-negative payoff of Yellow is greater than that of Red)

The objective function is of the form of equation 2 with

$$A = (1 - w(1 - P_R p_{XR})) \text{ and } B = (w(1 - P_R p_{XR}) - w(1 - P_R p_{XR} - P_Y p_{XY}))$$

So, the solution is $AL = \frac{EB^{-1/R}eY^{1-1/R}}{A^{-1/R}eR^{1-1/R} + B^{-1/R}eY^{1-1/R}}$ with A and B as above.

I think in the software we need to calculate both solutions and check the condition.

RDEU ambiguity

When subjects are confronted with and distorted pl_{XR} and ph_{XR} , they take the average of these distorted probabilities:

$$p_{XR} = \frac{w(pl_{XR}) + w(ph_{XR})}{2}$$

And then it reduces to the risk case, where we still need to consider the two solutions.

RDEU work It is quite similar with what we have done in EU. We use two sets (risk and ambiguity parts) to estimate parameters r and g , and then we use the estimated r and g to estimate np (the number of puzzles subjects thought they can solve correctly). We can use the same equations as above to calculate the optimal allocations in this case.

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